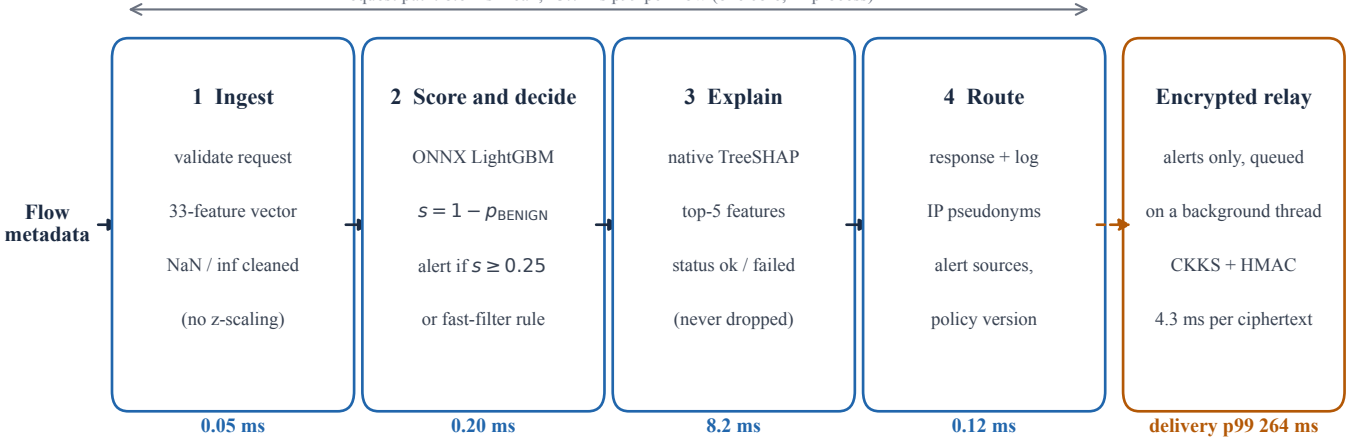

request path: 8.6 ms mean, 13.7 ms p99 per flow (one core, in-process)
Flow metadata
1 Ingest
validate request
33-feature vector
NaN / inf cleaned
(no z-scaling)
0.05 ms
2 Score and decide
ONNX LightGBM
$s = 1 - p_{\text{BENIGN}}$
alert if $s \geq 0.25$
or fast-filter rule
0.20 ms
3 Explain
native TreeSHAP
top-5 features
status ok / failed
(never dropped)
8.2 ms
4 Route
response + log
IP pseudonyms
alert sources,
policy version
0.12 ms
Encrypted relay
alerts only, queued
on a background thread
CKKS + HMAC
4.3 ms per ciphertext
delivery p99 264 ms